§ Notation / Introduction

$\mathcal{H} := \{ z \in \mathbb{C} \mid Im(z) > 0 \}$ $(x, y) = z = x + iy$

$GL_2\mathbb{R}^+ := \{ 2\times 2 \text{ Matrices, Positive determinant} \}$ (Multiplication)

$SL_2\mathbb{R} := \{ \quad " \quad \text{determinant } 1 \}$

$SO_2\mathbb{R} := \left\{ \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \mid 0 \le \theta < 2\pi \right\}$

Target.
↳ Undergraduate-level Review of hyperbolic geometry

A 3-dimension Manifold by identifying det. hence locally compact, T_2

• (σ-algebra of open set)

Note: • $SO_2\mathbb{R}$ is a compact subgroup of $SL_2\mathbb{R}$

• A closed subgroup is compact iff it is a conjugate of a conjugate of a closed subgroup of 2×2 orthogonal matrices.

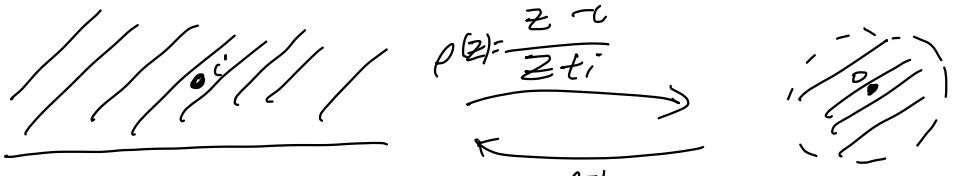
$SL_2\mathbb{R}$ Acts on $\mathcal{H}$: $\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2\mathbb{R}$ $z \in \mathcal{H}$

$\alpha \cdot z := \dfrac{az+b}{cz+d}$, (express · from now on)

The book uses $j(\alpha, z)$ to denote the denominator

Thm. $Stab_i = SO_2\mathbb{R}$

Pf. ⊇ obvious. take $\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in Stab_i$. $\rho : \mathcal{H} \to \mathbb{D}$



$f(z) = \dfrac{az+b}{cz+d}$. ↻ f (induced by α) such that $f(i) = i$

↻ $\mu := \rho f \rho^{-1}$

μ is an automorphism of $\mathbb{D}$ with $\mu(0) = 0$, so by Schwartz Lemma, μ is a rotation: $\mu(z) = e^{i\theta} z$.

Then, check that $f(z) = \rho^{-1}\mu\rho = \dfrac{(\cos\theta)z + \sin\theta}{(-\sin\theta)z + \cos\theta}$ ✳

§ Measure on $\mathbb{H}$ and $SL_2(\mathbb{R})$

Riemannian Metric: At Each point $\vec{p} = (x, y)$, an inner product.

$\begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix}_p$ dependent on p (locally), $\left(\begin{matrix}2\text{-dimensional} \\ \text{real manifold}\end{matrix}\right)$

For $\mathbb{H}$, the metric we take: $\vec{p}=(x,y) \longmapsto \begin{pmatrix} \frac{1}{y^2} & 0 \\ 0 & \frac{1}{y^2} \end{pmatrix}$

(In more manifold language: Riemannian metric $\frac{1}{y^2}(dx^2+dy^2)$)

With a metric: Do calculus!

- Length of Curve $\gamma:[0,1]\to\mathbb{H}$, $t \mapsto (x(t), y(t))$ $= \int_0^1 \sqrt{\langle ds, ds\rangle}$,

$$= \int_0^1 \frac{\sqrt{\left(\frac{dx(t)}{dt}\right)^2 + \left(\frac{dy(t)}{dt}\right)^2}}{y(t)}\, dt.$$

- Geodesics: (Shortest length curve between two points)
  Vertical Lines or Arcs centered at $\mathbb{R}$.

- Areas $\Omega \subseteq \mathbb{H}$, $\text{Area}(\Omega) = \boxed{\int_\Omega \frac{1}{y^2}\, dx\, dy.}$ ← Haar Measure.

Example: Triangle  has area $\pi - \theta_1 - \theta_2 - \theta_3$.

Thm. (Iwasawa Decomposition) any $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2\mathbb{R}$ can be uniquely written as:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y^{1/2} & 0 \\ 0 & y^{-1/2} \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

For some $x, y, \theta \in \mathbb{R}$ with $y>0$, $0 \le \theta < 2\pi$.

Cor. Identifying $(x,y)\in\mathbb{H}$, $SL_2\mathbb{R} \cong \mathbb{H} \times SO_2\mathbb{R}$ (as manifolds)

Pf: Given $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \alpha \in SL_2\mathbb{R}$, find $x, y \in \mathbb{R}$ s.t. $\alpha i := \frac{ai+b}{ci+d} = x + yi$.

Now the map $z \mapsto \frac{z-x}{y}$ map αi to i, so

$$\frac{1}{\sqrt{y}}\begin{pmatrix} 1 & -x \\ 0 & \sqrt{y} \end{pmatrix} \alpha\, i = i \quad \Rightarrow \quad \frac{1}{\sqrt{y}}\begin{pmatrix} 1 & -x \\ 0 & \sqrt{y} \end{pmatrix}\alpha \in \text{Stab}_i = SO_2.$$

$$\frac{1}{\sqrt{y}}\begin{pmatrix} 1 & -x \\ 0 & y \end{pmatrix} \alpha = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

∎

Now, we have coordinate (x, y, θ) in $SL_2\mathbb{R}$, then define.

Defn (Measure on $\mathbb{H}$) Given $\Omega \subseteq \mathbb{H}$, we define

$$\nu(\Omega) = \int_\Omega \frac{1}{y^2}\, dx\, dy$$

Defn: (Measure on $SL_2\mathbb{R}$) Given $\widetilde{\Omega} \subseteq SL_2\mathbb{R}$, we define.

$$\mu(\widetilde{\Omega}) = \int_{\widetilde{\Omega}} \frac{1}{2\pi y^2}\, dx\, dy\, d\theta$$

The key of having a measure is to integrate (weird) functions $f: \Omega \to \mathbb{C}$, we can do this on $\mathbb{H}$ and $SL_2\mathbb{R}$ now:

Prop. ① for any $\Omega \subseteq \mathbb{H}$, $\nu(\alpha\Omega) = \nu(\Omega)$

② For any $\widetilde{\Omega} \subseteq SL_2\mathbb{R}$ $\mu(\alpha\widetilde{\Omega}) = \mu(\widetilde{\Omega})$

③ For any $f: \mathbb{H} \to \mathbb{C}$, define $g(\alpha) := f(\alpha i)$, $g: SL_2\mathbb{R} \to \mathbb{C}$

Then we have $\int_{\mathbb{H}} f\, d\mu = \int_{SL_2\mathbb{R}} g\, d\nu$.

Proof: ① Ω with coordinate $(x, y) \in \mathbb{H}$, Now take $\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2\mathbb{R}$.

Let $\alpha\Omega$ have coordinate $(u, v) \in \mathbb{H}$, that means, u, v are functions of (x, y) [write $u(x,y)$, $v(x,y)$]. That is, $u + iv = \dfrac{a(x+iy)+b}{c(x+iy)+d}$

$$u(x,y) = \frac{ac(x^2+y^2) + (ad+bc)x + bd}{(cx+d)^2 + (cy)^2} \qquad v(x,y) = \frac{y}{(cx+d)^2 + (cy)^2} = \frac{y}{|cz+d|^2}$$

$$\nu(\alpha\Omega) = \int_{\alpha\Omega} \frac{du\, dv}{v^2} = \int_\Omega \det\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} \frac{|cz+d|^4}{y^2}\, dx\, dy.$$

$$= \int_\Omega \frac{1}{|cz+d|^4} \cdot \frac{|cz+d|^4}{y^2}\, dx\, dy = \int_\Omega \frac{1}{y^2}\, dx\, dy = \nu(\Omega)$$

② $\widetilde{\Omega}$ coordinate (x, y, θ), $\alpha\widetilde{\Omega}$ coordinate $(\tilde{u}, \tilde{v}, \phi)$ ✱

Def: Iwasawa decompose $\begin{pmatrix} a & b \\ c & d \end{pmatrix}\left[\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}\begin{pmatrix} y^{1/2} & 0 \\ 0 & y^{-1/2} \end{pmatrix}\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\right]$

to $\begin{pmatrix} 1 & \tilde{u} \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \tilde{v}^{1/2} & 0 \\ 0 & \tilde{v}^{-1/2} \end{pmatrix}\begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}$

Turns out: $\tilde{u} = u$, $\tilde{v} = v$, $\phi = \theta - \arg(cz+d) = \theta - \tan^{-1}\left(\frac{cy}{cx+d}\right)$

(functions of x, y, θ). Now

$$\mu(\alpha\Omega) = \int_{\alpha\Omega} \frac{d\tilde{u}\, d\tilde{v}\, d\phi}{2\pi\, \tilde{v}^2} = \int_\Omega \overbrace{\det\begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix}}^{J} \frac{dx\, dy\, d\theta\, |cz+d|^4}{2\pi\, y^2}$$

Note: $J = \det \begin{pmatrix} u_x & u_y & u_\theta \\ v_x & v_y & v_\theta \\ \phi_x & \phi_y & \phi_\theta \end{pmatrix} = \det \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}$ we calculated,

$\mu(\alpha\Lambda) = \int_{\Omega} \frac{dx\,dy\,d\theta}{2\pi\, y^2} = \mu(\Lambda)$ ∎

③ : Exercise.

Rmk. (P.16)

- They are Haar Measure on $\mathbb{H}$ and $SL_2\mathbb{R}$. they are unique up to $\times C$
- They are also "right invariant", meaning that they are unimodular.

§ Actions and Fuchsian Groups

$\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2\mathbb{R}$, characteristic equation $x^2 - tr(\alpha)x + 1 = 0$

Discriminant: $tr^2(\alpha) - 4$.

Defn. If $|tr(\alpha)| = \begin{cases} < 2 \\ = 2 \\ > 2 \end{cases}$ we call α an elliptic element.
" parabolic "
" hyperbolic "

Intuition: $z \mapsto \frac{az+b}{cz+d}$ also can be seen on $GL_2\mathbb{R} \circlearrowright \mathbb{P}^1$,

$z = \frac{az+b}{cz+d} \Leftrightarrow cz^2 + (d-a)z - b = 0 \Leftrightarrow$ discriminant $(d-a)^2 + 4bc = (d+a)^2 - 4$

∴ The Eigen value tell me the fixed points of the transformation.

Elliptic: Two complex conjugate, e.g. inversion $z \mapsto -\frac{1}{z}$ $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$
Parabolic: 1 real or ∞ fixed point, e.g. translation $z \mapsto z+3$ $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$
Hyperbolic: Two real or ∞ fixed point e.g. Scaling $z \mapsto 4z$ $\begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$

In fact...

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y^{1/2} & 0 \\ 0 & y^{-1/2} \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

Parabolic Hyperbolic Elliptic.

Rmk. The property is invariant under conjugation

Defn Γ is discrete if there is no limit point in Γ.

Defn. Γ acts properly discontinuously on X if for any compact $A, B \subseteq X$, $\#\{\gamma \in \Gamma \mid \gamma A \cap B \neq \emptyset\} < \infty$

Thm. Γ is discrete $\Leftrightarrow$ Γ acts properly discontinuously on X.

Defn. Γ is called a Fuchsian group if Γ is discrete.

Examples: $SL_2(\mathbb{Z})$, $\Gamma(n) = \left\{\gamma \in SL_2\mathbb{Z} \mid \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \bmod n\right\}$

Any subgroup of $SL_2(\mathbb{Z})$ is called a modular group.

Defn. A modular form of weight k with respect to modular group Γ of a Γ-form weight k is a meromorphic function $f: \mathbb{H} \to \mathbb{C}$ such that $\forall z \in \mathbb{H}$, $f(z) = (cz+d)^{-k} f\left(\frac{az+b}{cz+d}\right)$

for any $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$.

Next:
- ↳ Cusps, elliptic/hyperbolic/parabolic points on $\mathbb{H}$.
- ↳ Regular and Irregular Cusps
- ↳ Normalizers of groups "generated" by elements